

Non-Revenue Water Investigation

Ol Moran Water System — Kenya

Metering calibration review based on 3-day device log analysis · Prepared July 3, 2026

Executive Summary

Field observations show the eWater-monitored taps at Ol Moran are recording **10-20% less water than what is actually being dispensed** — a gap classified as Non-Revenue Water (NRW): water that leaves the system but is never metered or billed. This review analyzed device logs from the past 3 days across the 25 dispensing assets at Ol Moran that have an active preload setting (the eSense gateway asset and assets with no configured preload were excluded as they do not perform metered dispensing).

The data confirms the field observation is explainable by two device settings that control how raw flow-sensor ticks are converted into a litre reading: the **Litres Conversion Factor (LCF)** and the **Preload** offset. Both are currently configured with generic, largely uniform values across the network rather than values calibrated to each tap's actual hardware — and at more than half of the taps analyzed, this produces a measurable under-count relative to the expected 20 L typical dispense.

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Active taps under-metering vs. the 20 L benchmark

10.5%

Average under-count among under-metering taps

4

Taps under-metering in the reported 10-20% NRW band

348-392

Range of LCF values in use (should be one verified value per tap model)

Bottom line: the LCF and Preload settings are not tap-specific calibrations — they are close to network-wide defaults. Because the volume formula (below) divides directly by LCF, any tap whose real ticks-per-litre differs from its configured LCF will silently under- or over-report every single dispense by a fixed percentage, indefinitely, until recalibrated.

How the Metering Works — and Where It Breaks Down

Each dispense is measured by counting flow-sensor pulses ("ticks"). The device converts ticks to litres with:

$$\text{Recorded Volume (L)} = \text{Flow Ticks (FC)} \div \text{LCF (ticks per litre)}$$

An additional **Preload** value (in ticks) is added by the device firmware to compensate for a small, unavoidable amount of flow that occurs as the valve closes at the end of a dispense —

water that physically passes through the tap but occurs just outside the main counting window ("valve-close overrun"). This preload is meant to be set to each tap's actual measured overrun.

1. LCF miscalibration — the primary driver

LCF is a hardware property of each tap's flow sensor — it should be set once, from a physical calibration (dispense a known volume, count the ticks, compute ticks/litre). If the configured LCF is set higher than the tap's true ticks-per-litre, the formula above divides by too large a number, and every dispense is recorded as smaller than it really was — permanently, until the setting is corrected. This is a pure calibration error, not a hardware fault, and it scales linearly: a 10% error in LCF produces an almost exact 10% under-count of every litre that passes through the tap.

Using the 3-day dispense-volume distribution per tap (assuming the typical real-world dispense at these community taps is 20 L, per field guidance), we can back out what each tap's LCF should be for its recorded "typical dispense" to land on 20 L. The results show LCF values are inconsistent across the network (currently 348–392 as configured, but the implied "correct" values needed to hit 20 L range from 213 to 564 across different taps) — strong evidence that LCF has not been calibrated per-tap.

2. Preload under-compensation — a secondary, tap-specific driver

Nearly every tap in this network is running the same default Preload value (400 ticks, or 365/366 on a few), rather than a value derived from that tap's own measured overrun. The device logs let us directly observe the real overrun per tap (from "no-credit" events, which capture the unmetered ticks at valve-close). The measured values range enormously — from 39 ticks up to 442 ticks across different taps at Ol Moran — which means the same flat 400-tick preload is **too low** for some taps (under-compensating, adding to NRW) and **too high** for others (slightly over-compensating). At Ol Moran, most taps are currently over-compensated on preload (19 of 22), while 2 are under-compensated; preload is a smaller contributor than LCF here, but it moves in the same direction as LCF error whenever both are under-set, compounding the loss.

Why this specifically causes Non-Revenue Water: NRW is water that is produced/dispensed but not captured in billing or reporting. When LCF is set too high (or Preload too low), the physical tap dispenses the full volume the user pays for or draws, but the device under-reports it. The "missing" litres are real water that left the system — they simply never appear in the metered totals, which is the textbook definition of apparent-loss NRW caused by meter inaccuracy (as opposed to physical leakage).

Findings by Asset

The table below covers the 22 Ol Moran taps with sufficient 3-day dispense data (asset 2105, the eSense gateway, and 3 assets with no configured preload or no dispense activity in the window are excluded — see appendix). Positive "vs. 20 L" values indicate under-metering; negative values indicate over-metering.

Taps under-metering (13 of 22) — likely NRW contributors

Asset	ID	LCF (set)	Implied correct LCF	Typical dispense (L)	vs. 20 L	Preload (set)	Preload (measured)
Kwa David Wainaina	2241	389	213	10.95	+45.3%	400	182.3
Daniel Lomoe	2240	373	309	16.55	+17.2%	400	218.4
Kwa Ng'ang'a	2258	365	312	17.1	+14.5%	400	39
Edward Waiguru	2227	375	330	17.6	+12%	400	212
Community BH	2257	390	348	17.85	+10.7%	400	169.3
David Maina Mwangi	2254	362	330	18.25	+8.8%	400	354
Richard Esokon	2252	385	355	18.45	+7.8%	400	216.8
Stevin Njoroge	2226	382	360	18.85	+5.7%	400	161.6
Eliot Njoroge	2256	382	363	19	+5%	400	274.6
Deborah Wangari	2251	368	355	19.3	+3.5%	400	368.1
Wilfred Eyanae	2243	381	372	19.55	+2.2%	400	201.7
Eunice Muguri Kamau	2245	354	346	19.55	+2.2%	400	234.8
Mama Ataboh	2249	365	358	19.6	+2%	365	310.8

Taps over-metering (9 of 22) — LCF set too low for that tap

Asset	ID	LCF (set)	Implied correct LCF	Typical dispense (L)	vs. 20 L	Preload (set)	Preload (measured)
ACC	2250	392	564	28.8	-44%	400	—
Kwa David	2242	369	413	22.4	-12%	400	279.6
Kijito BH	2247	375	420	22.4	-12%	400	246
Robert Longole	2233	358	396	22.1	-10.5%	400	109
OL MORAN 1	2076	365	402	22.05	-10.3%	365	371
	2246	348	378	21.75	-8.7%	400	167.9

Nelson Zakayo
Mwangi

Zipporah Wairimo	2248	369	376	20.4	-2%	400	442
Kwa Mwai	2253	379	386	20.35	-1.8%	400	368
Kwa Penina Naputo	2259	381	388	20.35	-1.8%	400	261

"Implied correct LCF" = the LCF value that would make this tap's observed typical dispense equal 20 L, given its current data. It is a diagnostic estimate, not a replacement for a physical field calibration.

Recommended Actions

Action	Priority	Detail
Physically recalibrate LCF on the 13 under-metering taps	High	Dispense a known, measured volume (e.g. 20 L) at each tap and recompute ticks/litre directly, rather than trusting the "implied correct LCF" estimate. Prioritize Kwa David Wainaina (45% gap), Daniel Lomoe (17%), Kwa Ng'ang'a (15%), and Edward Waiguru (12%) — these show the largest and most confidently-estimated under-count.
Set Preload from each tap's measured overrun, not a network default	High	Update the Preload setting per tap to match its own measured no-credit overrun ticks (already visible in the device logs) instead of the current flat 400-tick default used across nearly the whole network.
Re-run this analysis after recalibration	Medium	Repeat the 3-day dispense-volume and preload comparison after settings are corrected to confirm the under-count has closed to within a few percent of 20 L, and to catch any tap that still drifts.
Investigate the 3 taps with zero dispense activity	Medium	OL MORAN 2, Benson Kamala, and Geoffrey Njoroge recorded no valid dispense events in 3 days — this may indicate a connectivity, wiring, or usage issue independent of calibration, and should be checked on-site.
Standardize a calibration procedure and schedule	Low	Once corrected, LCF and Preload should not need frequent changes, but a periodic (e.g. annual) physical spot-check would catch sensor drift before it re-accumulates into NRW.

Appendix — Assets Excluded from This Analysis

Asset	ID	Reason excluded
eSENSe OI Moran	2105	Gateway/sensor asset, not a dispensing tap
OL MORAN 2	2077	Preload configured, but zero dispense events in the 3-day window
Benson Kamala	2244	Preload configured, but zero dispense events in the 3-day window
Geoffrey Njoroge	2255	Preload configured, only 6 dispense events — insufficient for a reliable typical-dispense estimate

Methodology: flow-tick data was decoded from 39-byte EWC datalog packets received over the last 3 days per asset. Dispense volume = Flow Count (FC) ÷ LCF, filtered to the 10-30 L analysis window; "typical dispense" is the peak of the resulting distribution (KDE mode). Measured preload = average unmetered ticks captured on "no-credit" (valve-close) events. All figures are derived from live device telemetry via the eWater State API.